F79SU – SOLUTIONS – 2023-24

Maps to IFoA subject CS2 – *Risk Modelling and Survival Analysis.*

1. *This is an extension of tutorial-type questions. It additionally requires interpretation of Cox model in the context of a categorical variable with more than 2 levels. CS2 syllabus topic 4.2 (Estimation of lifetime distributions)*

(a) $\lambda_A(t) = \lambda_0(t)$; $\lambda_B(t) = \lambda_0(t)\exp(\beta_B)$; and $\lambda_C(t) = \lambda_0(t)\exp(\beta_C)$. [1 mark each]

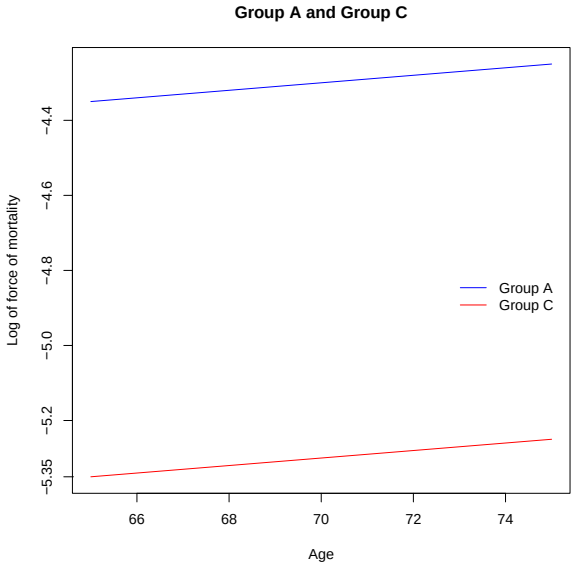
(b) (i) $\lambda_A(t) = \exp(a + bt) \Rightarrow \log(\lambda_A(t)) = -5 + 0.01t = -4.35$ at $t = 65 \Rightarrow \lambda_A(65) = 0.012907$. [1 mark]

(ii) $\lambda_B(t) = \exp(a + bt)\exp(\beta_B) \Rightarrow \log(\lambda_B(t)) = -5 + 0.01t + \beta_B = -4.35 - 0.85 = -5.2$ at $t = 65 \Rightarrow \lambda_B(65) = 0.005517$. [1 mark]

(iii) $\lambda_C(t) = \exp(a+bt)\exp(\beta_C) \Rightarrow \log(\lambda_C(t)) = -5+0.01t+\beta_C = -4.35-1 = -5.35$ at $t = 65 \Rightarrow \lambda_C(65) = 0.004748$. [1 mark]

(c) The graph will show t changing from 65 to 75 on the x-axis with an initial value identified in (i) and (iii). The hazard rate will increase wrt t where the difference between group A and group C will be constant by β_C over the time. [1 mark]
Thus, the main feature of the graph would be 'parallel curves'. [1 mark]

Figure 1: Log of force of mortality people aged between 65 and 75 in group A and group C



(d) The results suggest that an individual with higher education (group C) has a relatively lower risk of death compared to another person with lower education (group A). [1 mark]

2. (a) We have

$$
\begin{aligned}
{}_tp_x^{00} &= e^{-(\sigma+\mu)t} \\
{}_tp_x^{11} &= e^{-(\rho+\nu)t}.
\end{aligned}
$$

[1 mark each]

(b) (i)

$$
\begin{aligned}
L(\sigma,\mu,\rho,\nu;\mathbf{d},\mathbf{u},\mathbf{r},\mathbf{s},\mathbf{v},\mathbf{w}) &\propto \prod_{i=1}^{n} e^{-(\sigma+\mu)v_i}\mu^{d_i}\sigma^{s_i}e^{-(\rho+\nu)w_i}\rho^{u_i}\nu^{r_i} \\
&= e^{-(\sigma+\mu)\sum v_i}\mu^{\sum d_i}\sigma^{\sum s_i}e^{-(\rho+\nu)\sum w_i}\rho^{\sum u_i}\nu^{\sum r_i} \\
L(\sigma,\mu,\rho,\nu) &\propto e^{-(\sigma+\mu)v}\mu^{d}\sigma^{s}e^{-(\rho+\nu)w}\rho^{u}\nu^{r}
\end{aligned}
$$

[2 marks]

(ii) The maximum likelihood estimators of σ, μ, ρ, ν, respectively, are

$$
\begin{aligned}
\hat{\sigma} &= \frac{200}{8000} = 0.025, \\
\hat{\mu} &= \frac{300}{8000} = 0.0375, \\
\hat{\rho} &= \frac{30}{1500} = 0.02, \\
\hat{\nu} &= \frac{100}{1500} \approx 0.0667.
\end{aligned}
$$

[3 marks]

(iii) The approximate standard errors are given by

$$
\begin{aligned}
SD(\hat{\mu}) &\approx \sqrt{\hat{\mu}/v} = \sqrt{0.0375/8000} = 0.002165, \\
SD(\hat{\rho}) &\approx \sqrt{\hat{\rho}/w} = \sqrt{0.02/1500} = 0.003651, \\
SD(\hat{\nu}) &\approx \sqrt{\hat{\nu}/w} = \sqrt{0.0667/1500} = 0.006667,
\end{aligned}
$$

[1 mark]

giving approximate 95% confidence intervals of

$$
\begin{aligned}
&(\hat{\mu} - 1.96 * SD(\hat{\mu}), \hat{\mu} - 1.96 * SD(\hat{\mu}) = (0.0333, 0.0417), \\
&(\hat{\rho} - 1.96 * SD(\hat{\rho}), \hat{\rho} - 1.96 * SD(\hat{\rho}) = (0.0128, 0.0272), \\
&(\hat{\nu} - 1.96 * SD(\hat{\nu}), \hat{\nu} - 1.96 * SD(\hat{\nu}) = (0.0536, 0.0797).
\end{aligned}
$$

[2 marks]

We observe that none of the confidence intervals overlap suggesting somewhat different death rates. In particular, those with the critical illness die at a far higher rate than healthy individuals, and those with the critical illness are unlikely to die of other causes. [1 mark]

3. *Similar examples to all parts seen in class and tutorials. For parts (d) and (e), students need to apply the Wald test learned in other chapters on a new setting. CS2 syllabus topic 4.3 (Derive MLEs for transition intensities)*

(a) We can compute the likelihood contribution for each life using the following table:

Life	q_i	d_i	Contribution
1	$1-\exp(-\mu/2)$	1	$1-\exp(-\mu/2)$
2	$1-\exp(-0.2\mu)$	0	$\exp(-0.2\mu)$
3	$1-\exp(-0.8\mu)$	0	$\exp(-0.8\mu)$
4	$1-\exp(-\mu/4)$	1	$1-\exp(-\mu/4)$

[2 mark]

Therefore, the likelihood function is

$$\begin{aligned} L(\mu) &= \left(1-e^{-\mu/2}\right) e^{-0.2\mu} e^{-0.8\mu} \left(1-e^{-\mu/4}\right) \\ &= e^{-\mu} \left(1-e^{-\mu/2}\right) \left(1-e^{-\mu/4}\right). \end{aligned}$$

[1 mark]

(b) The log-likelihood is

$$l(\mu) = -\mu + \log\left(1-e^{-\mu/2}\right) + \log\left(1-e^{-\mu/4}\right),$$

[1 mark]

and the score function is

$$\begin{aligned} U(\mu) := \frac{dl(\mu)}{d\mu} &= -1 + \frac{0.5e^{-\mu/2}}{1-e^{-\mu/2}} + \frac{0.25e^{-\mu/4}}{1-e^{-\mu/4}} \\ &= -1 + \frac{0.5}{e^{\mu/2}-1} + \frac{0.25}{e^{\mu/4}-1}. \end{aligned}$$

[1 mark]

Since $U(1.4966) = 0.0000310987 \approx 0$, it follows that $\widehat{\mu} = 1.4966$ is the MLE. [1 mark]

(c) Under the assumption of constant force of mortality and using the invariance property of MLEs,

$$\widehat{q}_{50} = 1 - e^{-\widehat{\mu}} = 1 - e^{-1.4966} = 0.7761.$$

[1 mark]

(d) By the Delta method, we have

$$\text{Var}\left(\widehat{q}_{50}\right) \approx e^{-2\widehat{\mu}}\text{Var}\left(\widehat{\mu}\right) = 0.05012677 * 0.4564 = 0.0229.$$

[1 mark]

Then the Wald test statistic is

$$z^2 = \frac{\left(\widehat{q}_{50} - 0.5\right)^2}{\text{Var}\left(\widehat{q}_{50}\right)} = \frac{\left(0.7761 - 0.5\right)^2}{0.0229} = 3.33.$$

[1 mark]

(e) Under H_0, z^2 is approximately χ^2-distributed with 1 degree of freedom. From the table, the critical value is 3.841. Since z^2 is smaller, this implies that we do not have enough evidence to reject H_0 at $\alpha = 0.05$. [1 mark]

4. *Similar examples to all parts seen in class and tutorials. Parts (a), (b)(i) will test students' ability to critically comment on graduation.*
CS2 syllabus topic 4.5 (Graduation and tests)

(a) For setting premium rates mortality must be a smooth function of age. This is not the case with $\hat{\mu}_x$ (e.g. see age 67), and therefore we need to graduate the crude estimates. [2 marks]

(b) (i) Model looks reasonably suitable. However, discrepancy between $\hat{\mu}_x$ and $\dot{\mu}_x$ looks somewhat high for some ages (e.g. at $x = 67, 70$). [1 mark]

(ii) Use χ^2 test for overall goodness-of-fit. We have:
$X = \sum_x z_x^2 = 1.265$, on $11 - 2 = 9$ degrees of freedom. [1 mark]
Critical value from tables is $\chi^2_{9,0.05} = 16.92$. [1 mark]
Therefore, there is *no* evidence against the hypothesis that the fit of the graduation is good. [1 mark]

(iii) We have $n_1 = 5$ positive signs, $n_2 = 6$ negative signs and $g = 4$ groups of positive signs. [1 mark]
From tables (e.g. 'yellow' book), the largest value g^* for which we would reject the null hypothesis (that positive and negative residuals are in random order) at the 5% level, is $g^* = 1$. Alternatively, exact p-value is given by

$$P(G \leq 4) = \sum_{t=1}^{4} \frac{\binom{n_1-1}{t-1}\binom{n_2+1}{t}}{\binom{n_1+n_2}{n_1}} = 0.955.$$

Therefore, there is *no* evidence against the null hypothesis.
[1 mark]
The test is checking for 'local' bias in the graduation. [1 mark]

(iv) Approximately, $\hat{\mu}_x \sim N\left(\mu_x, \frac{\hat{\mu}_x}{E_x^c}\right)$. [1 mark]
Therefore, an approximate 95% CI for μ_{70} will be given by

$$\begin{aligned} 0.0396 \pm 1.96\sqrt{\frac{0.0396}{808}} &= 0.0396 \pm 1.96 \times \sqrt{4.901 \times 10^{-5}} \\ &= 0.0396 \pm 1.96 \times 0.007 \\ &= 0.0396 \pm 0.01372 \end{aligned}$$

i.e. $(0.0259, 0.0533)$. [2 marks]
Now, $\dot{\mu}_{70} = \exp(-10.3595 + 0.1013 \times 70) = 0.0381$, and therefore the two estimates are consistent. [2 marks]

5. *Students have seen similar examples implemented in R. Calculating projected values manually will test their understanding. Parts (c) and (d) will further test their understanding of projection in the future and the modelling behind it.*
CS2 syllabus topic 4.6 (Mortality projection)

(a)

$$\hat{\kappa}_{2025} = \hat{\kappa}_{2023} + 2 * \hat{\delta} = -29.3 + 2 * (-0.543) = -30.386.$$

[1 mark]

$$\hat{\kappa}_{2025} \pm 2.5758 * \sqrt{2} * \sqrt{0.892} = (-33.8264, -26.9456).$$

[3 marks]

(b)

$$\log(\mu_{70.5,2025.5}) = \hat{\alpha}_{70} + \hat{\beta}_{70} * \hat{\kappa}_{2025} = -3.9 + 0.018 * (-30.386) = -4.446948$$

[1 mark]

(c) The prediction interval will now be considerably wider, as it depends on $\sqrt{m}\sigma$, where m is the number of years from the last year in the past data. [1 mark]

This reflects higher uncertainty as the projection moves further away into the future. [1 mark]

(d) The new model uses age as a numerical variable and has considerably fewer parameters compared to the earlier model (only one α parameter, compared to 31 earlier). [1 mark]

This will be useful if the data support a linear smooth function for the age (α) component of the model. [1 mark]

The fit of the models can be compared by examining the residuals against age. [1 mark]

6. *Similar examples to all parts seen in class and tutorials.*
CS2 syllabus topic 4.4 (Principle of correspondence; Exposed to risk)

(a) Calendar year rate interval, from 1/7 of each year to 30/6 of following year. [1 mark]

(b) We have

$$P^D_{x,t} = P^C_{x+1,t}.$$

[1 mark]

Therefore, $P^D_{70,0} = P^C_{71,0} = 6878$. [1 mark]

Because of calendar year rate interval, all lives aged 70 on 1/7/17 will change to age label 71 on 1/7/18. [1 mark]

(c) At the start of the rate interval exact age is between x and $x+1$, and assuming that births are uniformly distributed throughout the year, average age will be $x + \frac{1}{2}$. Therefore, $\hat{\mu}_{70}$ applies to exact age 71 and $\hat{q}_{70}$ applies to exact age 70.5. [2 marks]

[END OF SOLUTIONS]